

IAS Architecture

ANISHA M. LAL

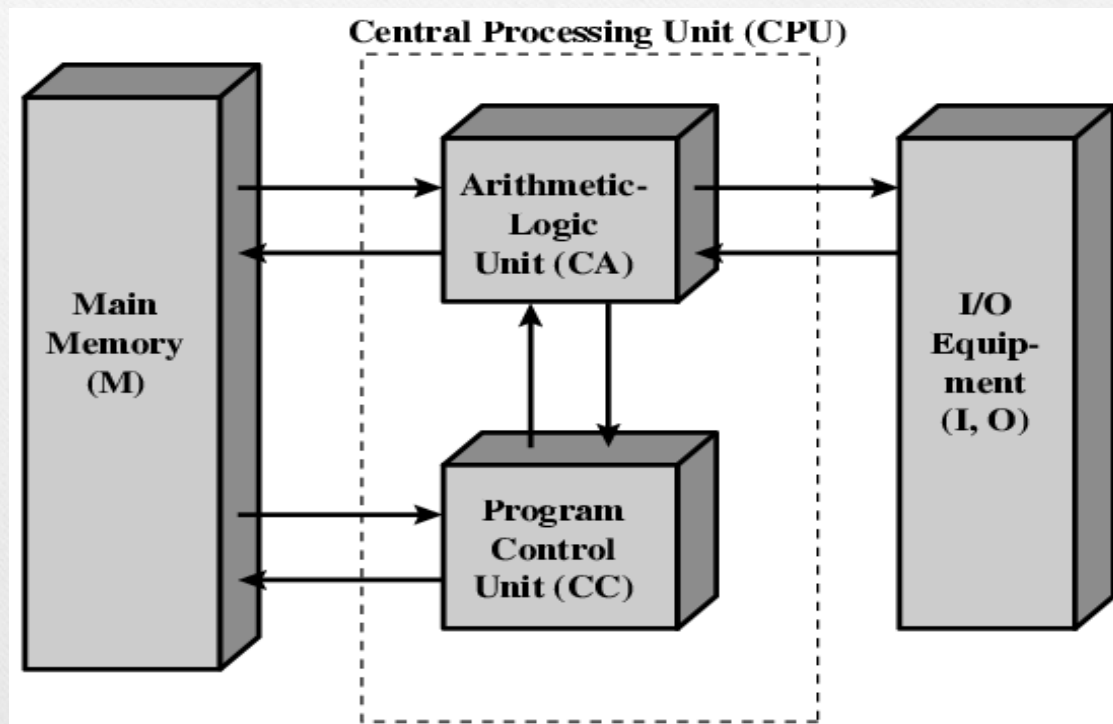
Von Neumann Machine

- Stored Program concept:
 - The task of entering and altering programs for the ENIAC was extremely tedious.
 - The programming process could be facilitated if the program could be presented in a form suitable for storing in memory alongside the data.
 - Then a computer could get its instructions by reading them from memory, and a program could be set or altered by setting the values of a portion of memory.
 - The stored program concept idea is given by a mathematician John Von Neumann, who was the consultant on the ENIAC project.

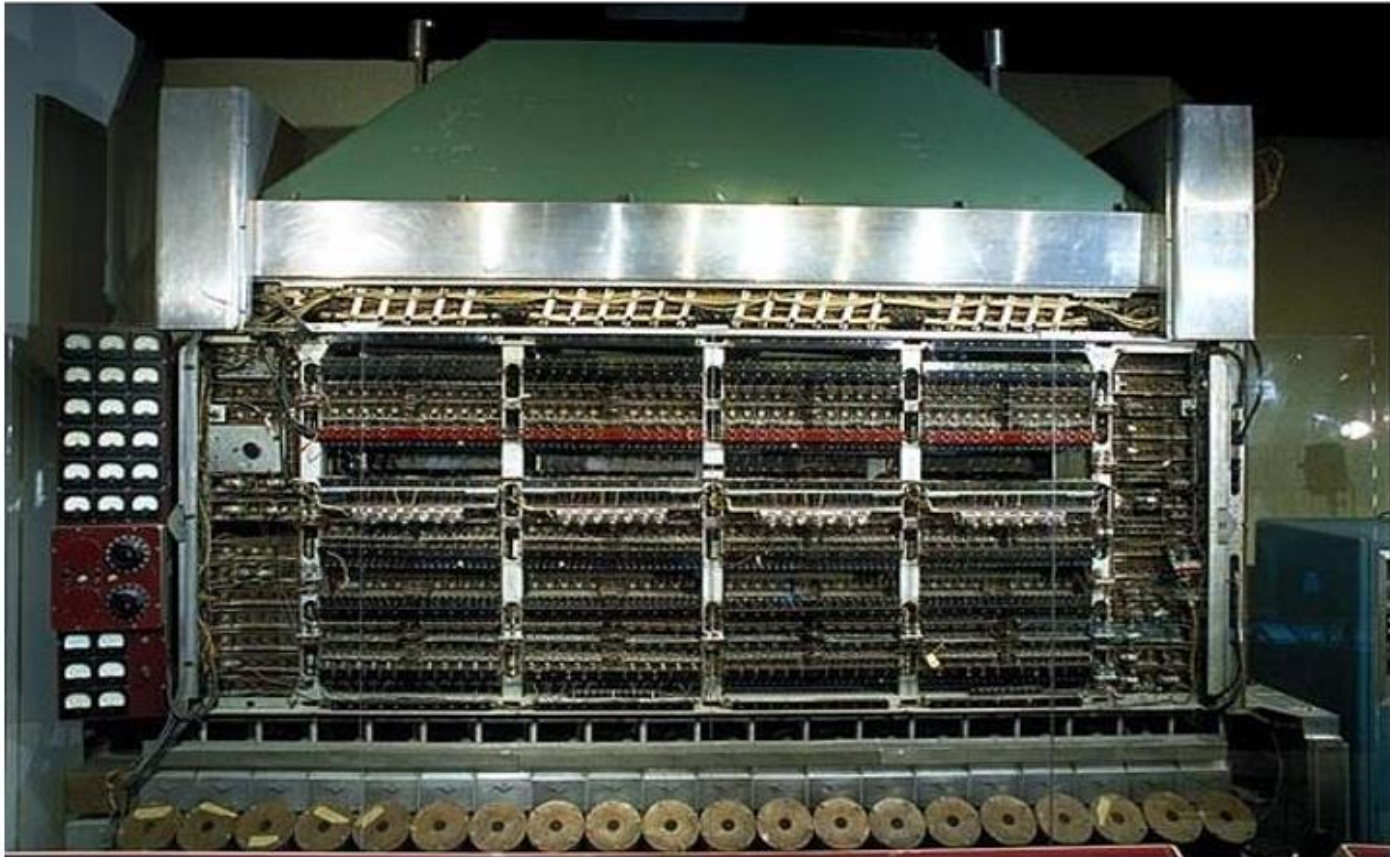
Von Neumann machine

IAS – Institute of Advanced Studies

A prototype developed by John Von Neumann in 1946 at Princeton University.



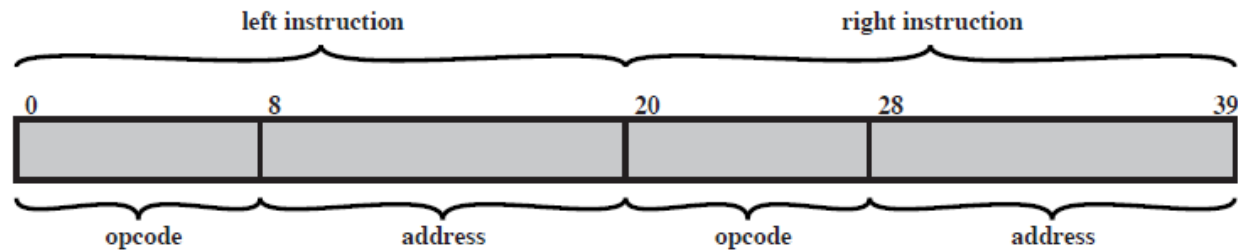
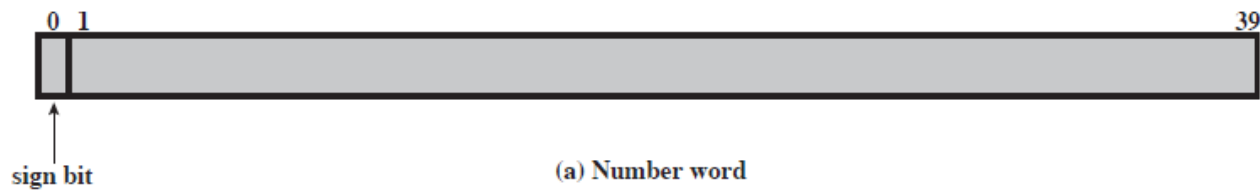
The IAS Computer, 1952



IAS – memory format

- 1000 x 40 bit words (1000 storage locations of 40 binary bits each)
- **Data Word Format:**
 - Each number is represented by a sign bit and a 39 bit value.
- **Instruction word Format:**
 - A word may contain two 20 bit instructions.
 - Each instruction consisting of :
 - an 8 bit operation code (opcode) specifying the operation to be performed
 - a 12 bit address designating one of the words in memory (0 to 999)

IAS – memory format



Structure of IAS Machine

– Memory Address Register (MAR)

- Specifies the address in memory of the word to be written from or read into the MBR.

– Instruction Register (IR)

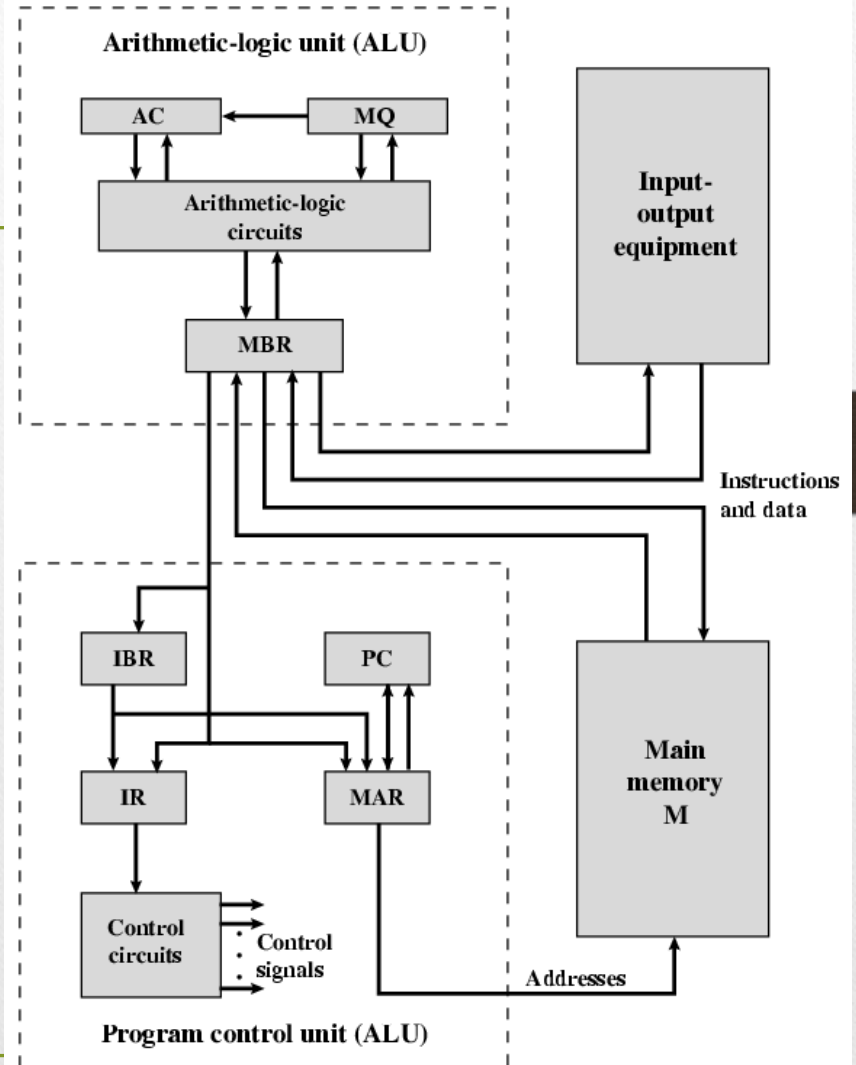
- Contains the 8 bit opcode instruction being executed.

– Instruction Buffer Register (IBR)

- Employed to hold temporarily the right hand instruction from a word in memory

– Program Counter (PC)

- Contains the address of the next instruction pair to be fetched from memory



Registers in IAS Machine

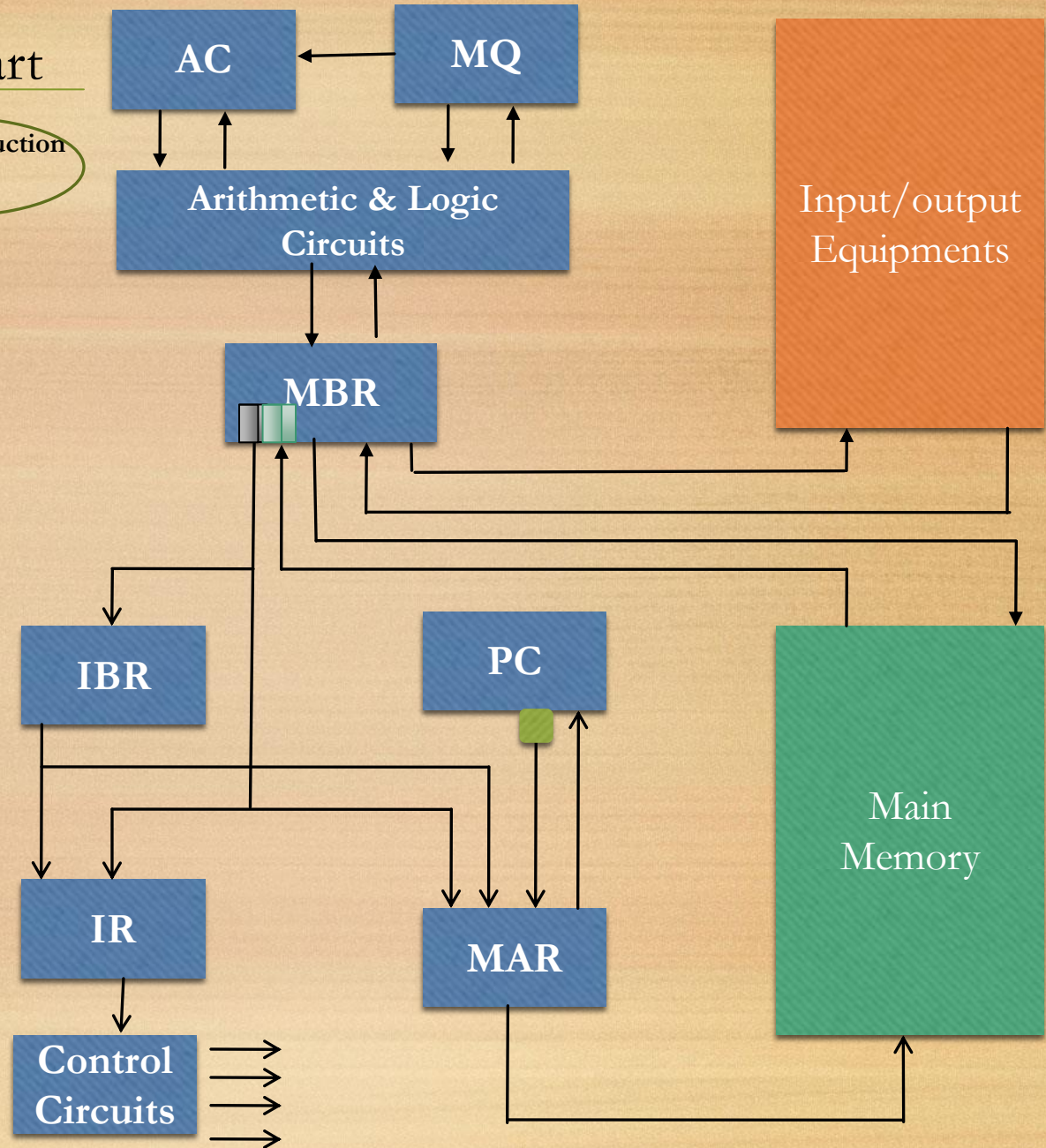
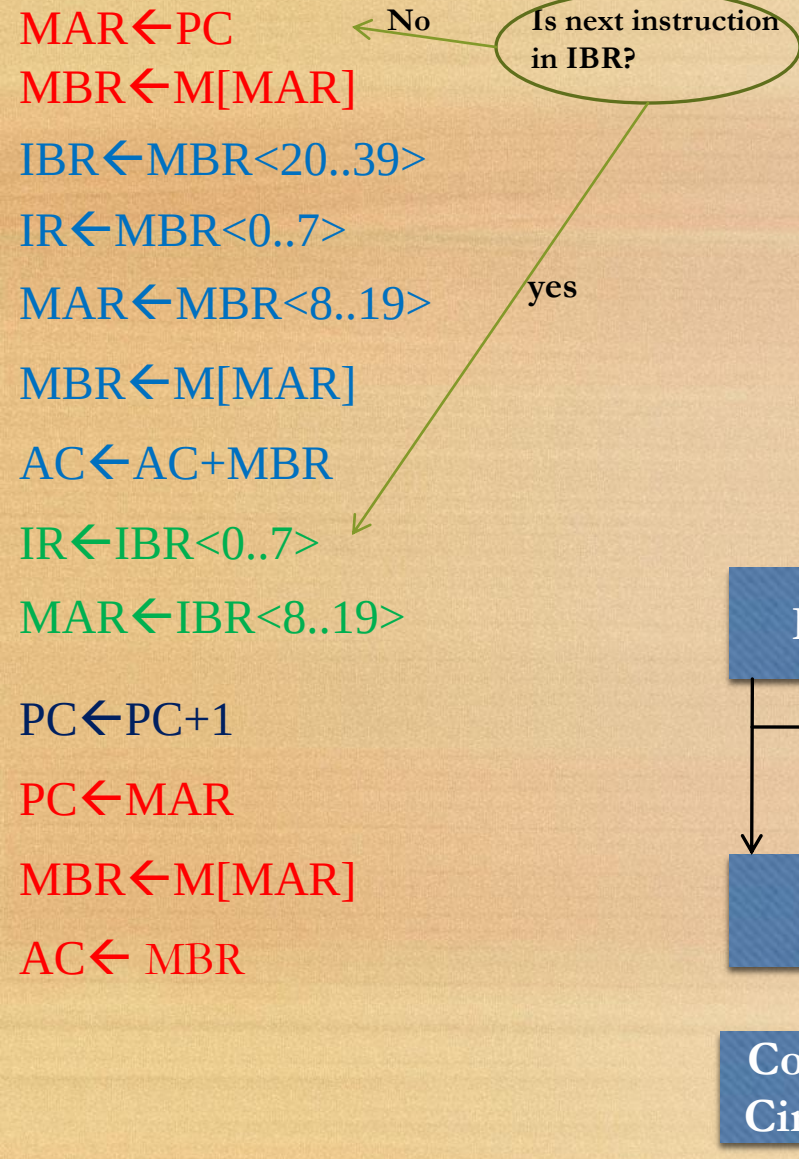
– Memory Buffer Register (MBR)

- Contains a word to be stored in memory or sent to the I/O unit, or it is used to receive a word from memory or from the I/O unit.

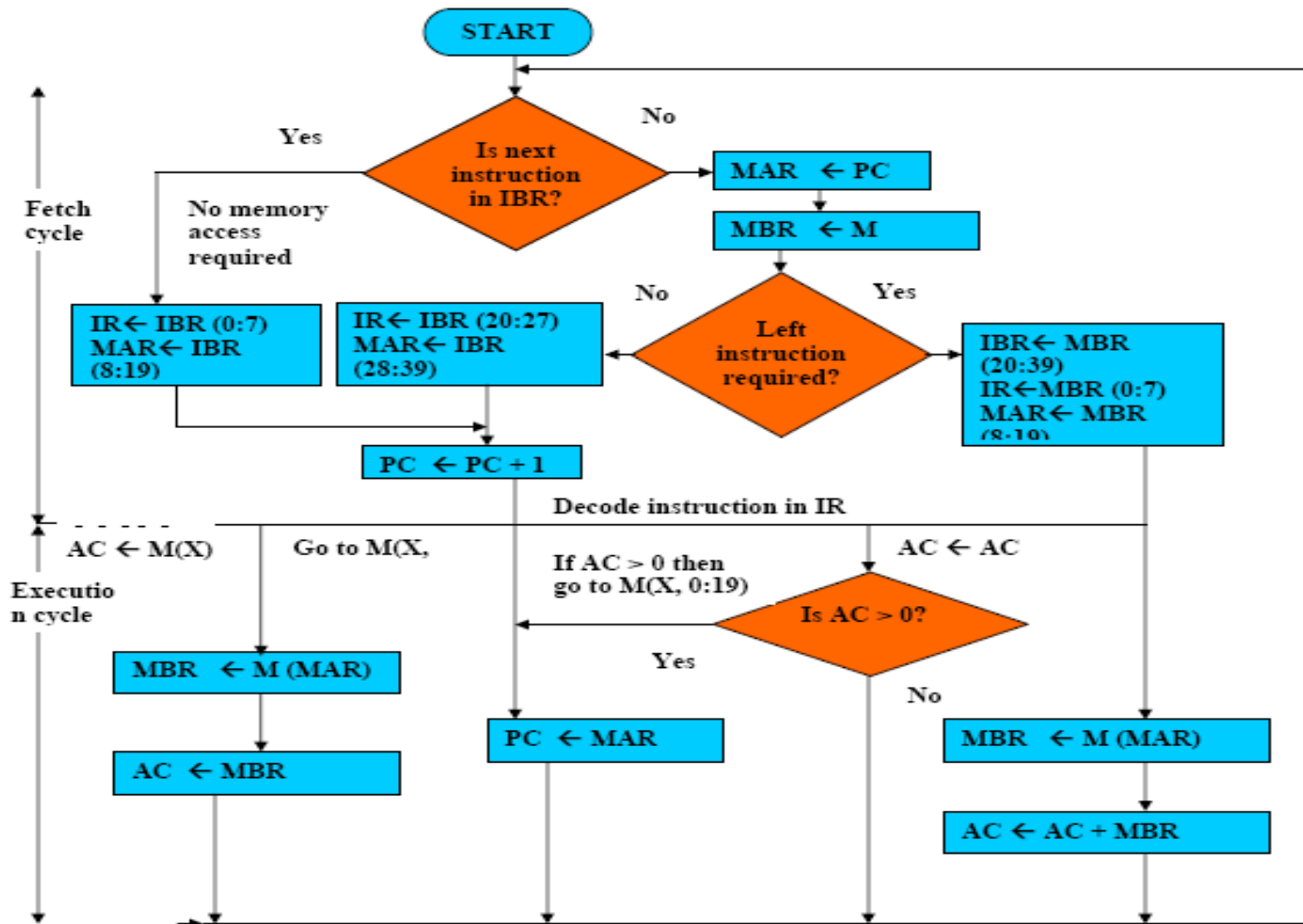
– Accumulator (AC) & Multiplier Quotient (MQ)

- Employed to hold temporarily the output data. For eg. The result of multiplying two 40 bit numbers is an 80 bit number, the most significant 40 bits are stored in the AC and the least significant in the MQ.

IAS Operation - Flowchart



Fetch / Execute Cycle



IAS Instructions

IAS – Total 21 Instructions

- Data Transfer
- Unconditional Branch Instruction
- Conditional Branch Instruction
- Arithmetic
- Address Modify Instruction

IAS Instruction set

Instruction Type	Opcode	Symbolic Representation	Description
Data transfer	00001010	LOAD MQ	Transfer contents of register MQ to the accumulator AC
	00001001	LOAD MQ,M(X)	Transfer contents of memory location X to MQ
	00100001	STOR M(X)	Transfer contents of accumulator to memory location X
	00000001	LOAD M(X)	Transfer M(X) to the accumulator
	00000010	LOAD -M(X)	Transfer -M(X) to the accumulator
	00000011	LOAD M(X)	Transfer absolute value of M(X) to the accumulator
	00000100	LOAD - M(X)	Transfer - M(X) to the accumulator
Unconditional branch	00001101	JUMP M(X,0:19)	Take next instruction from left half of M(X)
	00001110	JUMP M(X,20:39)	Take next instruction from right half of M(X)
Conditional branch	00001111	JUMP+ M(X,0:19)	If number in the accumulator is nonnegative, take next instruction from left half of M(X)
	00010000	JUMP+ M(X,20:39)	If number in the accumulator is nonnegative, take next instruction from right half of M(X)

IAS Instruction set Contd..

Arithmetic	00000101	ADD M(X)	Add M(X) to AC; put the result in AC
	00000111	ADD M(X)	Add M(X) to AC; put the result in AC
	00000110	SUB M(X)	Subtract M(X) from AC; put the result in AC
	00001000	SUB M(X)	Subtract M(X) from AC; put the remainder in AC
	00001011	MUL M(X)	Multiply M(X) by MQ; put most significant bits of result in AC, put least significant bits in MQ
	00001100	DIV M(X)	Divide AC by M(X); put the quotient in MQ and the remainder in AC
	00010100	LSH	Multiply accumulator by 2, i.e., shift left one bit position
	00010101	RSH	Divide accumulator by 2, i.e., shift right one position
Address modify	00010010	STOR M(X,8:19)	Replace left address field at M(X) by 12 rightmost bits of AC
	00010011	STOR M(X,28:39)	Replace right address field at M(X) by 12 rightmost bits of AC